

Chapter-05

1. Autokey Cipher

The Autokey Cipher is a polyalphabetic substitution cipher. It is very similar to the Vigenère Cipher, but the method of generating the key is different. Instead of repeating the keyword again and again, the Autokey Cipher uses the plaintext itself to extend the key. This makes the key stream less repetitive and improves security compared to the standard Vigenère Cipher.

- More secure than the ordinary Vigenère Cipher.
- However, it can still be attacked using frequency analysis if enough ciphertext is available.

How the Autokey Cipher Works

Suppose the keyword is only one letter:

N

Plaintext:

HELLO

Instead of repeating the key:

NNNNN

the key is extended using the plaintext itself.

Key stream becomes: N

H E L L

Notice that the last plaintext letter (**O**) is not used because only five key letters are needed. **Encryption**

Formula

$$C_i = (P_i + K_i) \bmod 26$$

Where

- P_i = Plaintext letter
- K_i = Key stream letter
- C_i = Ciphertext letter

Decryption Formula

$$P_i = (C_i - K_i) \bmod 26$$

Alphabet to Number Conversion

Letter	Number
A	0
B	1
C	2
D	3
E	4
F	5

G	6
H	7
I	8
J	9
K	10
L	11
M	12
N	13
O	14
P	15
Q	16
R	17
S	18
T	19
U	20
V	21
W	22
X	23
Y	24
Z	25

Example

Plaintext

HELLO

Initial Key

N

Key Stream

N H E L L

Convert into numbers

Plaintext H E L L O

Number 7 4 11 11 14

Key **N H E L L**

Number 13 7 4 11 11

Now add both numbers

7+13 4+7 11+4 11+11 14+11

20 11 15 22 25

Convert back to letters

20 11 15 22 25

U L P W Z

Final Ciphertext

ULPWZ

Decryption Example

Ciphertext:

ULPWZ

Initial Key:

N

Key Stream:

N H E L L

Convert into Numbers

Ciphertext U L P W Z

Number 20 11 15 22 25

Key N H E L L

Number 13 7 4 11 11

Now Subtract the Key Values (Modulo 26)

Calculation 20-13 11-7 15-4 22-11 25-11

Result 7 4 11 11 14

(If any result is negative, add 26 before taking modulo 26.)

Convert Back to Letters

Number 7 4 11 11 14

Letter H E L L O

Recovered Plaintext

HELLO

Advantages

- More secure than the ordinary Vigenère Cipher.
- Does not repeat the keyword continuously.
- Makes frequency analysis more difficult.

Disadvantages

- Still vulnerable to frequency analysis.
- If part of the plaintext is known, the remaining key stream may be discovered.

2. Vernam Cipher

The Vernam Cipher is a polyalphabetic substitution cipher that uses a key whose length is equal to the length of the plaintext. Each plaintext letter is combined with the corresponding key letter to produce the ciphertext.

- Originally proposed using a very long repeating key.
- The ideal version, using a truly random non-repeating key, is called the **One-Time Pad**.

Main Idea

Unlike the Vigenère Cipher, the key is not repeated. Instead, the key length must be exactly the same as the plaintext length.

Example

Plaintext

HELLO

Key

XMCKL

Both contain five letters.

Character Numbering

Assign numbers to letters.

A = 0

B = 1

...

Z = 25

Encryption Algorithm Step 1

Convert the plaintext letters into numbers.

Step 2

Convert the key letters into numbers.

Step 3

Add the corresponding numbers.

Step 4

If the result is **26 or greater**, subtract **26** (equivalent to taking modulo 26).

Step 5

Convert the resulting numbers back into letters.

Encryption Formula

$$C = (P + K) \bmod 26$$

Where

- **P** = Plaintext number
- **K** = Key number
- **C** = Ciphertext number

Decryption Formula

$$P = (C - K) \bmod 26$$

Vernam Cipher Example

Given:

Plaintext:

RAMSWARUPK

Key:

RANCHOBABA

Step 1: Convert Letters into Numbers

Using **A = 0, B = 1, ..., Z = 25**

Plaintext	R	A	M	S	W	A	R	U	P	K
Number	17	0	12	18	22	0	17	20	15	10
Key	R	A	N	C	H	O	B	A	B	A
Number	17	0	13	2	7	14	1	0	1	0

Step 2: Add Plaintext and Key Values

$$C = (P + K) \bmod 26$$

Calculation:

17+17	0+0	12+13	18+2	22+7	0+14	17+1	20+0	15+1	10+0
34	0	25	20	29	14	18	20	16	10

Step 3: Apply Modulo 26

Since **34** and **29** are greater than 25, subtract 26:

- $34 \bmod 26 = 8$
- $29 \bmod 26 = 3$

Final numbers:

8 0 25 20 3 14 18 20 16 10

Step 4: Convert Numbers Back to Letters

Number **8 0 25 20 3 14 18 20 16 10**

Letter **I A Z U D O S U Q K**

Final Ciphertext

IAZUDOSUQK

Decryption

Reverse the encryption process. Subtract the key numbers from the ciphertext numbers using modulo 26 to recover the original plaintext.

Vernam Cipher Decryption Example

Given:

Ciphertext:

IAZUDOSUQK

Key:

RANCHOBABA

Step 1: Convert Letters into Numbers

Using **A = 0, B = 1, ..., Z = 25**

Ciphertext	I	A	Z	U	D	O	S	U	Q	K
Number	8	0	25	20	3	14	18	20	16	10
Key	R	A	N	C	H	O	B	A	B	A
Number	17	0	13	2	7	14	1	0	1	0

Step 2: Subtract Key Values from Ciphertext Values

$$P = (C - K) \bmod 26$$

Calculation:

8-17	0-0	25-13	20-2	3-7	14-14	18-1	20-0	16-1	10-0
-9	0	12	18	-4	0	17	20	15	10

Step 3: Apply Modulo 26

For negative values, add **26**:

- $-9 + 26 = 17$
- $-4 + 26 = 22$

Final numbers:

17 0 12 18 22 0 17 20 15 10

Step 4: Convert Numbers Back to Letters

Number	17	0	12	18	22	0	17	20	15	10
Letter	R	A	M	S	W	A	R	U	P	K

Recovered Plaintext

RAMSWARUPK

Advantages

- Stronger than the Vigenère Cipher when the key is not repeated.
- Easy to implement.
- Uses a different key character for every plaintext character.

Disadvantages

- Requires a key as long as the message.
- Key distribution is difficult.
- Reusing the same key weakens security.

3. One-Time Pad (OTP)

The One-Time Pad (OTP) is an unconditionally secure encryption technique that uses a random secret key with the same length as the plaintext. Each key is used only once and then discarded. It provides perfect secrecy, meaning the ciphertext cannot be broken even with unlimited computing power.

It uses:

- A truly random key.
- A key equal in length to the plaintext.
- A key that is used only once.

Under these conditions, the cipher is theoretically unbreakable because the ciphertext has no statistical relationship to the plaintext.

Conditions for Perfect Security

A One-Time Pad is perfectly secure only if:

1. The key is truly random.
2. The key length equals the message length.
3. The key is kept secret.
4. The key is used only once.

Why is OTP Unbreakable?

For any given ciphertext, there are many possible plaintexts, each with a different key. Without the correct random key, an attacker cannot determine which plaintext is the real one.

Problems with OTP

Although OTP is perfectly secure, it is rarely used because:

- Generating truly random keys is difficult.
- Securely distributing long keys is difficult.
- Every key can be used only once.

Vernam Cipher vs One-Time Pad

Vernam Cipher	One-Time Pad
Key length equals plaintext length	Key length equals plaintext length
Key may repeat or follow a pattern	Key is truly random
May not be perfectly secure	Perfectly secure (theoretical)
Key may be reused	Key must never be reused

4. Monoalphabetic Cipher vs Polyalphabetic Cipher

Feature	Monoalphabetic Cipher	Polyalphabetic Cipher
Substitution Alphabet	Uses only one fixed substitution alphabet.	Uses multiple substitution alphabets.
Letter Mapping	A plaintext letter always maps to the same ciphertext letter.	A plaintext letter can map to different ciphertext letters.
Relationship	Has a one-to-one relationship between plaintext and ciphertext.	Has a one-to-many relationship between plaintext and ciphertext.
Security	Less secure because letter patterns and frequencies are preserved.	More secure because letter frequencies are hidden better.
Cryptanalysis	Easier to break using frequency analysis.	Harder to break due to changing substitution patterns.
Examples	Caesar Cipher, Affine Cipher, Monoalphabetic Substitution.	Vigenère Cipher, Autokey, Playfair, Hill Cipher, One-Time Pad.